

When Refusal Fails Quietly: Hallucination Analysis of a RAG Chatbot for Philippine Solid Waste Law

Ram Kelly Jersen Leliza
College of Computer Studies and Engineering
Jose Rizal University
Manila City, Philippines
ramkellyjersen.leliza@my.jru.edu

Abstract—Philippine solid waste management is governed by Republic Act No. 9003 and its implementing rules, whose provisions on segregation and penalties are dispersed across two lengthy documents. This paper presents a domain-specific retrieval-augmented generation chatbot built over that corpus and evaluates its grounding under adversarial configuration. The documents were chunked at 500 characters, embedded using all-MiniLM-L6-v2 and indexed in ChromaDB, with generation constrained by a context-only system prompt. An out-of-domain fallback test and a stress-test raising the decoding temperature and removing the fallback instruction were conducted. Grounding survived every out-of-domain sample but failed for an in-domain boundary query, where the system produced a confidently misattributed statutory penalty.

Keywords—*retrieval-augmented generation, hallucination, prompt constraints, vector database, solid waste management*

I. SYSTEM ARCHITECTURE

A. Domain, Scope and Policy Alignment

The chatbot answers questions on the legal requirements governing how solid waste is classified, segregated, collected and disposed of in the Philippines, together with the institutional responsibilities and penalties attached to them. The scope is national rather than municipal.

The domain aligns with Sustainable Development Goal 11, Target 11.6, on reducing the adverse per-capita environmental impact of cities with attention to municipal waste management, and with Goal 12, Target 12.5, on substantially reducing waste generation through prevention, reduction, recycling and reuse [7].

Republic Act No. 9003 establishes a mandatory nationwide segregation regime across sixty-six sections [1], and its implementing rules add the operational detail the statute delegates to the Department [2]. The information most often needed in practice is dispersed and expressed in statutory language. A general-purpose model queried on these matters may return a fluent but unverifiable answer, since it cannot indicate which provision supports its response. In a compliance domain an incorrect statement about penalties is not merely unhelpful but actively misleading.

B. Pipeline Stages

Ingestion uses a directory loader that delegates parsing to a document partitioner. A recursive character splitter then divides the text, descending through a hierarchy of separators to preserve semantic boundaries. Whole documents cannot be embedded effectively, since one vector over a long document averages away the detail that makes retrieval precise; the overlap keeps a statement split across a boundary recoverable from either side.

Each chunk is embedded by all-MiniLM-L6-v2 [6] into a 384-dimensional space where semantic similarity

corresponds to geometric proximity, and the vectors are indexed in an in-memory ChromaDB collection. Retrieval is semantic rather than lexical: a query in ordinary speech can match statutory text using the term segregation at source without sharing a keyword. The retrieved chunks are concatenated into the context placeholder of the system prompt and dispatched to the model, which returns both the answer and the documents that produced it, the traceability on which Section III depends.

II. IMPLEMENTATION DETAILS

TABLE I. PIPELINE CONFIGURATION PARAMETERS

Parameter	Value
Source documents loaded	2 (RA 9003; DAO 2001-34)
Chunks after splitting	726
Chunk size / overlap	500 / 50 characters
Embedding model	all-MiniLM-L6-v2 (384 dimensions)
Vector store	ChromaDB (in-memory)
Retrieval depth (top-k)	3
Generation model	openai/gpt-oss-20b (Groq)
Temperature: baseline / stress-test	0 / 1.0

Three deviations from the laboratory template were required. The specified model llama-3.1-8b-instant was deprecated and withdrawn from free-tier access, returning a model-not-found error; the provider names openai/gpt-oss-20b as its replacement [4], confirmed against the model-listing endpoint before adoption. The chain-construction functions were relocated out of the core framework package in its version 1.0 release and are now imported from the compatibility package [3], with signatures unchanged, and parsing required the PDF extra of the partitioner package, whose base installation raises an ImportError on PDF input [5]. For Task 3 the template directs that the model cell be edited in place; this implementation instead builds a second parallel chain in dedicated cells, so both configurations execute in one sequential run and the notebook stays reproducible. The credential is captured at runtime through a masked prompt and written only into the process environment, never echoed to cell output, written to disk, or persisted in the saved notebook. A safeguard was added after identical chunks were found filling two of three retrieval slots: the vector store appends rather than replaces when its construction cell is re-executed in a live session, so an assertion now confirms one embedding per chunk before any query runs.

III. EXPERIMENTAL RESULTS AND HALLUCINATION ANALYSIS

A. Baseline and Fallback Behaviour

Under the baseline configuration the in-domain query returned a summary of facility record-keeping, monitoring and signage obligations. Every claim was traceable to the retrieved text, and retrieval drew from both documents,

confirming the index behaves as one unified collection rather than a per-file search.

The out-of-domain query returned the mandated fallback string verbatim, yet three chunks were still retrieved, none bearing on the question. The retriever applies no similarity threshold, so context is always supplied regardless of relevance. Retrieval did not produce the refusal; the system prompt did.

TABLE II. TEST QUERIES AND OBSERVED RESPONSES

Query	Config.	Expected Response	Actual Model Output
Q1	Baseline	Summary grounded in retrieved context	Summary of record-keeping, monitoring and signage duties; all claims traceable
Q2	Baseline	Fallback message	I cannot answer based on the provided domain data. (exact match)
Q2	Ungrounded	Refusal or unsupported answer	I am sorry, but I cannot answer that. (refusal, non-standard wording)
Q3	Baseline	Fallback message	I cannot answer based on the provided domain data. (exact match)
Q3	Ungrounded	Refusal, as no provision applies to the stated actor	Placed the household under Sec. 48 pars. (4) to (7); stated P1,000 to P3,000

a. Q1: Summarize the key information found in the documents. Q2: What is the orbital period of Jupiter around the Sun? Q3: What is the exact fine in pesos for a household that fails to segregate its waste?

B. Ungrounded Configuration

The stress-test raised the temperature to 1.0 and removed the fallback instruction, retaining the context-only instruction, so any drift is attributable specifically to the loss of a defined refusal path. Because sampling at this temperature is not deterministic, the query was issued four times, and all four observations are reported in the notebook.

Every sample refused, and none reproduced the mandated fallback string: grounding was preserved while format compliance was lost entirely. No two samples were worded alike, confirming stochastic decoding was active. Two of the four explicitly referred to the provided context, the last naming the missing subject outright, which shows the context-only constraint remained in force after the fallback line was deleted. String matching on the fallback message therefore cannot be relied upon: downstream logic detecting refusal by comparison against the mandated string would have failed on every sample.

C. In-Domain Boundary Probe

The out-of-domain query tests only the trivial case, where the subject is absent and refusal is unambiguous. A supplementary test beyond the template probed the harder condition. Query Q3 asks for a specific penalty applied to a specific actor. The corpus contains adjacent material, so retrieval returns relevant chunks carrying peso figures, but no provision prescribes a household-specific fine for failing to segregate.

The baseline refused correctly. The ungrounded configuration did not: it placed the household under Section 48 paragraphs (4) to (7), reported the range of P1,000 to P3,000, and prefaced the claim with an appeal to the provisions supplied to it. Section and range are both present in the retrieved text and individually accurate. The attribution is not. Those paragraphs penalise causing or permitting the collection of non-segregated waste [1], conduct directed at

the collector rather than the household generating it, and nothing in the retrieved context links a household's failure to segregate to them. The model supplied the connection while presenting it as sourced.

This is plausible interpolation rather than fabrication. No invented statute or figure appears; a real provision and a real range were applied to an actor they do not cover, and the claim was framed as drawn from the supplied provisions. Every element is verifiable except the one that matters, and the appeal to the source makes the error harder to detect. Grounding failed not where the subject was absent, but where it was close.

IV. CONCLUSION

The safety of this system rests on three contributions that are not interchangeable, each narrower than it first appears. Prompt restrictions divide into two functions. The instruction to answer using only the provided context carried the grounding behaviour, surviving the removal of every other control across every out-of-domain sample. The fallback instruction governed output predictability rather than grounding, and its removal cost format compliance at once. Treating the two as one guardrail would misattribute the protection each provides.

Retrieval-based information proved necessary but not sufficient. By returning evidence alongside the answer it makes verification possible; the boundary probe would have been indistinguishable from a correct answer without the retrieved chunks. Retrieval alone offers no protection, returning nearest neighbours regardless of relevance and unable to signal that a question is unanswerable. Its weakest point is the adjacent subject, where relevant context invites an answer the source does not support.

Basic API-key protection contributes on a different axis. Confining the credential to the process environment prevents the most common route by which prototype notebooks leak keys, namely sharing a file in which the key is visible in a cell. A system whose credential is exposed is not trustworthy however well it is grounded. For a domain aligned with SDG targets 11.6 and 12.5, a fluent misattribution of a statutory penalty is more damaging than a refusal. Constraints should be evaluated individually rather than as one guardrail, and evaluation should include queries sitting near the edge of the corpus rather than outside it.

REFERENCES

- [1] Congress of the Philippines, Republic Act No. 9003: Ecological Solid Waste Management Act of 2000. Environmental Management Bureau, Department of Environment and Natural Resources. [Online]. Available: <https://pepp.emb.gov.ph/wp-content/uploads/2016/06/RA-9003-Ecological-Solid-Waste-Management-Act-of-2000.pdf>
- [2] Department of Environment and Natural Resources, DENR Administrative Order No. 2001-34: Implementing Rules and Regulations of Republic Act No. 9003, Dec. 2001. [Online]. Available: <https://emb.gov.ph/wp-content/uploads/2015/12/DAO-2001-34.pdf>
- [3] LangChain, LangChain v1 migration guide. [Online]. Available: <https://docs.langchain.com/oss/python/migrate/langchain-v1>
- [4] Groq, Model deprecation. [Online]. Available: <https://console.groq.com/docs/deprecations>
- [5] Unstructured, Full installation. [Online]. Available: <https://docs.unstructured.io/open-source/installation/full-installation>
- [6] Sentence-Transformers, all-MiniLM-L6-v2 model card. [Online]. Available: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
- [7] United Nations, The 17 Goals. Department of Economic and Social Affairs. [Online]. Available: <https://sdgs.un.org/goals>